CS 168
Spring 2020

Introduction to the Internet
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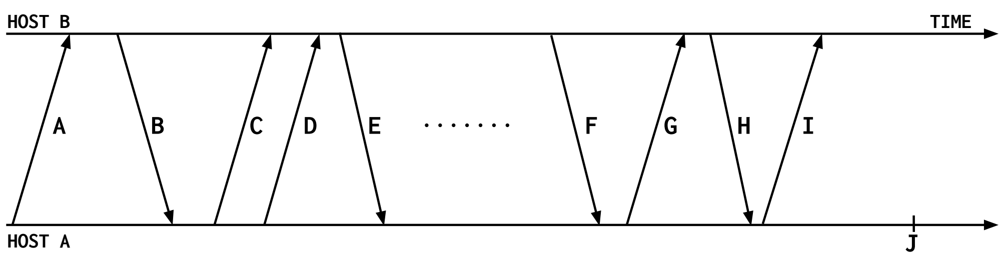
Discussion 9: TCP

1 Pop Quiz: TCP vs UDP

For each statement below, mark whether describes TCP, UDP, both (mark both) or neither (mark none).

	TCP	**UDP**
Provides reliable transport		
Limits messages to a single packet		
Has source port in the header		
Requires connection establishment		
You will use this for your project 2.		
It is optional to use the L4 checksum with this protocol (under IPv4)		

2 Flags



The above figure shows the life cycle of a TCP connection with normal termination - that is, connection establishment, data exchange, and teardown.

(1) For each of the arrows, choose whether it is a SYN, ACK, data, FIN or RST packet. A single arrow might have more than one of these flags set.

A: ___________ D: ___________ G: ___________

B: ___________ E: ___________ H: ___________

C: ___________ F: ___________ I: ___________

(2) When host A sends packet I, it sets a timer that ends at point J. What is the purpose of this timeout?

3 TCP Calculations

In this question, quantities will be measured in bytes unless **explicitly** mentioned otherwise.

(1) Suppose two hosts are about to open a TCP connection. The TCP headers used in the communication are only 20 bytes long and regular (no-options) IPv4 is being used for Layer 3. If the MTU of the link is 1260 bytes, what is the MSS?

(2) When this connection starts, the sender starts with an ISN 19. The initial window for the sender is set to 10 packets. Given the previously calculated MSS, what is the first ACK the sender receives and the last ACK the sender receives for this initial window? (Assume no packets were lost or reordered).

(3) In this part of the question we will determine the estimated RTT using some parameters of TCP. The following equations *may* be useful. Assume that connections have been open and that there are **no** dropped packets.

$$ETO = Estimated\_RTT + 4 \cdot Estimated\_Deviation$$
$$Estimated\_Deviation = \beta \cdot Estimated\_Deviation + (1-\beta) \cdot |Estimated\_RTT - Sample\_RTT|$$
$$Estimated\_RTT = \alpha \cdot Estimated\_RTT + (1-\alpha) \cdot Sampled\_RTT$$

a. The sender receives the current ACK and proceeds to update its various estimations. The time from when the packet was sent to when the ACK was received was $10msec$. If the previous $Estimated\_RTT$ was $70msec$, what will the new $Estimated\_RTT$ be (using $\alpha = 0.5$)?

b. If the previous $Estimated\_Deviation$ was $50msec$, what is the new $Estimated\_Deviation$ (using $\beta = 0.5$)?

c. Based on the previous 2 questions, what will the ETO be now?

(4) What is the maximum theoretical rate of data transfer for this window size if the $Estimated\_RTT$ is what was previously estimated?

(5) Assume $40msec$ is the $Estimated\_RTT$. If the lowest bandwidth across this connection is $76.25MBps$, what is the smallest window that optimizes the question?

4 TCP in Action

Consider a sender sending 1000 B of data to a receiver over **TCP**. The sender sends packets of 100B, the window size is 300B, and the ISN is 99 (so D100 is the first packet sent, then D200, and so on). Remember, TCP uses a sliding window, and retransmits the packet containing the next expected byte on a timeout or the 3rd duplicate ACK.

The link is **flaky**! The **initial** transmission of packets **D200 and D700** get dropped.

(1) Fill in the below table with all packets sent by the sender until the receiver has received all packets and the sender knows that. For simplicity, assume that packets (data and ACKs) arrive in order. You may or may not need to fill in all lines.

#	Packet # Sent	Sent on timeout/3rd duplicate ACK?	Dropped?	*Cumulative* ACK
1	D100			A200
2	D200		X	
3	D300			A200
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				

(2) If the RTT of the link is 10ms and the timeout is initially 3 seconds, what is the total time needed for the receiver to receive all packets and for the sender to know that? Assume small packets (negligible transmission delay) and negligible processing time, and that the estimates that go into the RTO remain constant during the events below.